Energy absorption

Rate of energy absorption in AC field

$$H = H(t) = H_0 + H'(t) \Rightarrow E(t) = \mathrm{Tr}\big(\rho(t)\, H(t)\big) \text{ is not conserved!}$$

$$\frac{dE}{dt} = \mathrm{Tr}\left[\rho(t)\frac{dH}{dt}\right] + \mathrm{Tr}\left[\frac{d\rho}{dt} H\right]$$

But $\frac{d\rho}{dt} = -i[H(t), \rho(t)] \Rightarrow \mathrm{Tr}\big\{[H,\rho]H\big\} =$

$$= \mathrm{Tr}\{H\rho H - \rho H H\} = \mathrm{Tr}\{HH\rho\} - \mathrm{Tr}\{HH\rho\} = 0$$

$$\Rightarrow \frac{dE}{dt} = \mathrm{Tr}\left\{\rho(t)\frac{dH}{dt}\right\}$$

Consider monochromatic perturbation

$$H'(t) = \hat{O}\, h e^{i\omega t} + \hat{O}^+ h^* e^{-i\omega t}$$

where h, h^* are complex amplitudes. Note $H'(t) = (H'(t))^+$

Then $\frac{dH}{dt} = i\omega\left(\hat{O} h e^{i\omega t} - \hat{O}^+ h^* e^{-i\omega t}\right)$

$$\Rightarrow \frac{dE}{dt} = i\omega\Big(\underbrace{\mathrm{Tr}\{\rho(t)\hat{O}\}}_{\langle \hat{O}(t)\rangle} h e^{i\omega t} - \underbrace{\mathrm{Tr}\{\rho(t)\hat{O}^+\}}_{\langle \hat{O}^+(t)\rangle} h^* e^{-i\omega t}\Big).$$

Linear response tells us

$$\langle \hat{O}(t)\rangle = \int_{-\infty}^{t} dt' \left(-i\langle[\hat{O}(t), \hat{O}(t')\, h e^{i\omega t'} + \hat{O}^+(t')\, h^* e^{-i\omega t'}]\rangle\right)$$

$$= \int_{-\infty}^{t} dt' \left(\chi_{OO}(t-t')\, h e^{i\omega t'} + \chi_{OO^+}(t-t')\, h^* e^{-i\omega t'}\right)$$

It is customary to calculate energy absorption over the period $T = \frac{2\pi}{\omega}$

$$\overline{\frac{dE}{dt}} = \frac{1}{T}\int_0^T dt\, \frac{dE}{dt} = \frac{1}{T}\int_0^{2\pi/\omega} dt\, (i\omega)\left\{ \langle \hat{O}(t)\rangle h e^{i\omega t} - \langle \hat{O}^+(t)\rangle h^* e^{-i\omega t} \right\} =$$

$$= \frac{i\omega}{T}\int_0^{2\pi/\omega} dt \left\{ \int_{-\infty}^{t} dt' \left(\chi_{OO}(t-t') h e^{i\omega t'} + \chi_{OO^+}(t-t') h^* e^{-i\omega t'} \right) \cdot h e^{i\omega t} - \int_{-\infty}^{t} dt' \left(\chi_{O^+O}(t-t') h e^{i\omega t'} + \chi_{O^+O^+}(t-t') h^* e^{-i\omega t'} \right) \cdot h^* e^{-i\omega t} \right\}$$

Change $t' = t - u \Rightarrow \int_{-\infty}^{t} dt'\, \chi(t-t')\, e^{\pm i\omega t'} =$

$$= \int_0^{\infty} du\, \chi(u)\, e^{\pm i\omega(t-u)}$$

$$\overline{\frac{dE}{dt}} = \frac{i\omega}{T}\int_0^{2\pi/\omega} dt \left\{ \int_0^{\infty} du \left(\chi_{OO}(u)\, h e^{i\omega(t-u)} + h^* e^{-i\omega(t-u)} \chi_{OO^+}(u) \right) \cdot h e^{i\omega t} - \int du \left(\chi_{O^+O}(u)\, h e^{i\omega(t-u)} + \chi_{O^+O^+}(u) e^{-i\omega(t-u)} \right) h^* e^{-i\omega t} \right\}$$

But $\frac{1}{T}\int_0^{2\pi/\omega} dt\, e^{i2\omega t} = \frac{1}{T\cdot i2\omega}\left(e^{i2\omega\frac{2\pi}{\omega}} - 1\right) = 0$; $\frac{1}{T}\int_0^{2\pi/\omega} dt = 1.$

So that $\overline{\frac{dE}{dt}} = i\omega \int_0^\infty du \left(\chi_{00+}(u) e^{i\omega u} h^* h \right.$

$$\left. - \chi_{0+0}(u) e^{-i\omega u} h h^* \right) = |h|^2 i\omega \left(\chi_{00+}(\omega) - \chi_{0+0}(-\omega) \right)$$

Here $\chi_{00+}(\omega) = \int_0^\infty du\, e^{i\omega u} \chi_{00+}(u) =$

$$= -i \int_0^\infty du\, e^{i\omega u} \langle [\hat{O}(u), \hat{O}^+] \rangle$$

Then $\chi_{0+0}(-\omega) = -i \int_0^\infty du\, e^{-i\omega u} \langle [\hat{O}^+(u), \hat{O}] \rangle$

Hence $\cancel{\left(\chi_{0+0}(-\omega)\right)^* = i \int_0^\infty du\, e^{i\omega u} \langle [\hat{O}^+(u), \hat{O}] \rangle^*}$

$\cancel{= i \int du\, e^{i\omega u} \langle \rho_0 [e^{iHu} \hat{O}^+ e^{-iHu}, \hat{O}] \rangle^*}$

$$\left(\chi_{00+}(\omega)\right)^* = i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 [e^{iHu} \hat{O} e^{-iHu}, \hat{O}^+] \rangle^*$$

$$= i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 [\hat{O}(u), \hat{O}^+]^+ \rangle =$$

$$= i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 \left(\hat{O}(u) \hat{O}^+ - \hat{O}^+ \hat{O}(u) \right)^+ \rangle =$$

$$= i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 \left(\hat{O} \hat{O}^+(u) - \hat{O}^+(u) \hat{O} \right) \rangle =$$

$$= -i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 [\hat{O}^+(u), \hat{O}] \rangle$$

$$= \chi_{0+0}(-\omega)$$

Thus $\overline{\frac{dE}{dt}} = |h|^2(i\omega)\ 2i\ \mathrm{Im}(\chi_{00+}(\omega))$

$$= -2\omega\ \mathrm{Im}(\chi_{00+}(\omega)) \cdot |h|^2.$$

Rate of energy absorption is determined by Im part of the susceptibility!

This is manifestation of FDT (fluctuation-dissipation theorem).